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Performance Analysis of ELA-CNN model for Image Forgery Detection

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***Abstract*—With the growth of technology, the amount of content being shared on day-to-day basis is a figure beyond our comprehension. This data can be shared in the form of photos and videos and can be spread using social media platforms like WhatsApp and Instagram. With this also comes the threat of forged images being used to share false information. Advanced tools and software have made it easy and quick to generate such images. Due to this, researchers have been focusing on devising methods to tackle the same in a hope to assist people in finding the authenticity of questionable images. Copy-move and splicing are two forms of commonly used forgery techniques. When a region of the image is copied and pasted on the same image it is referred to as copy-image forgery whereas a region of a different image is copied and pasted on another image it is said to be spliced forgery. Our technique makes use of error level analysis (ELA) that points out the difference between the compression ratios of authentic and manipulated image and hence highlights the latter. The extracted information is then fed into a variant of CNN which classifies the image as original or forged. We also provide a comparative analysis of the proposed method with statistical and deep learning models and it will be tested on the CASIA V2, MICC-F220 and MICC-F2000 forged image datasets.**

***Index Terms*—Image, Forgery, ELA, CNN, Deep Learning, Statistical methods, CASIA v2.0, MICC-F220, MICC-F2000.**

I. Introduction

According to a recent article by The Conversation, "If you use social media, the chances are you see (and forward) some of the more than 3.2 billion images and 720,000 hours of video shared daily." And the question posed by them was a rather important one – "When faced with such a glut of content, how can we know what's real and what's not?" Fake images have become a nuisance to our society. They can be used to spread false information, political and economic propaganda, sensationalizing of news and much more. They can prove to be especially dangerous during sensitive times where they can be used to aggravate mass emotions, examples of which we read in our newspapers from time to time. With the staggering amount of content that we consume everyday it becomes the responsibility of the consumers and the provider to differentiate between original and fake stuff. Due to software advancements deepfakes and spliced images have become fairly common. Tools, such as Photoshop2, GIMP3 make the job of manufacturing them simple and quick. Digital image forgery can be widely classified as – active image forgery which includes watermarking, and passive image forgery which entails splicing and copy-move forgery. To identify passive image forgery, extensive research has been conducted as they are of particular interest to us from the perspective of image forensics. Tackling this issue is a tricky task and it has been of prime importance to many researchers. Solutions proposed in the past include use of traditional ML algorithms while others make use of deep learning techniques. In this paper we propose a model that makes use of ELA – error level analysis and a variation of CNN – convolutional neural network. Error level analysis is a forensic technique that differentiates between the authentic and manipulated portion of the image based on the compression ratios. When an image is falsified, for example a region of the image is copied and pasted on another patch of the same image, there will be signification difference between the compression of the tampered region and the surrounding portion, whereas in a pristine image the compression should be roughly the same throughout the image. The ELA of the images is then fed into a special convolutional neural network, working of which will be elaborated further. CNN is a deep neural network that is widely used in fields such as image classification and image manipulation detection due to its ability to extract and learn the features of an image directly. It is mostly used where the input has a grid type data. We have summarized our contributions in the paper as follows:

- We propose a novel architecture which provides better accuracy over other methods.
- Compare the working of our proposed technique with both statistical and deep learning techniques.
- We compare the models on different datasets to test their functionality across a varied sample set.

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II. Related Work

Hakimi et al. [1] proposed a methodology where the chromatic components of an image were used to enhance the process of forgery detection. Upon testing on the CASIA v2.0 dataset they found out that the performance of the Cb component was almost similar to that of the Cr component which proved the consistency of their proposed approach. The method introduced in [2] aimed at reducing the amount of information that was processed to detect the forged images. It also ensured a more powerful block representation and this was achieved by using discrete wavelet transform (DWT) along with SVD. A technique based on LBP, tested on the COMOFOD dataset was developed by Mahale, V. H et al. [3] which was used to detect inconsistencies in the image. Forgery methods to investigate images stemming from watermarking and alpha matters were summarized by Wu-Chi et al. [4]

[5] suggested a method of copy-move manipulation detection using CNN. It was one of the first studies which introduced the idea of using the CNN algorithm for the purpose of manipulation detection in images and led the way for further research in this direction. This method however did not prove to be robust enough to be used in real-world scenarios.

III. Datasets

We have used three datasets for the purpose of this research namely – CASIA V2.0 [6], MICC-F220, MICC-F2000 [7]. The datasets have been handpicked to ensure that the proposed methodology is thoroughly tested on various use cases. The tampered images in these datasets are more realistic to the human eye and are difficult to catch with naked eye. The motivation behind selecting the dataset was that the techniques are tested across a myriad of images and the results are thorough taking into account most of the use cases present and prove to be useful across them all. The information about the datasets is summarized in **Table 1**.

IV. Proposed Methodology

A. Preprocessing

The following techniques are used in preprocessing:

- Image resizing: Resizing input images to a fixed size of (224,224,3)
- Image normalization: Scaling pixel values to a fixed range improves convergence during training and accuracy during testing.
- Color space conversion: Converting RGB to a different color space enhances relevant features for forgery detection.
- Error Level Analysis (ELA): Detecting regions with different compression levels using ELA identifies potential forgery regions.
- Image augmentation: Applying random transformations to input images generates additional training data and reduces overfitting.

B. ELA – Error Level Analysis

One of the most important techniques which used to identify the tampering of images is ELA [8]. Image compression is said to be a lossy process and ELA captures the compression distortion during the same. One such format where it proves to be useful is JPEG. When an image is taken using a digital camera it goes through the compression process once when we save the image . But when someone attempts to edit or tamper the image using software like Adobe Photoshop, GIMP, etc., the image goes through another round of compression. ELA captures the difference between the compression levels by calculating the ratio after storing them at a certain quality and locates the tampering. The tampered regions are represented by the local maxima whereas the local minima represents the authentic portions. When the images are saved as JPEG, it is done in blocks of 8 x 8 which go through individual compression. If an image is pristine the compression difference of all the 8 x 8 blocks remains similar across the entire image and as mentioned above it is irregular in case of a modified one. The ELA decreases further as the number of times the image is edited increases.

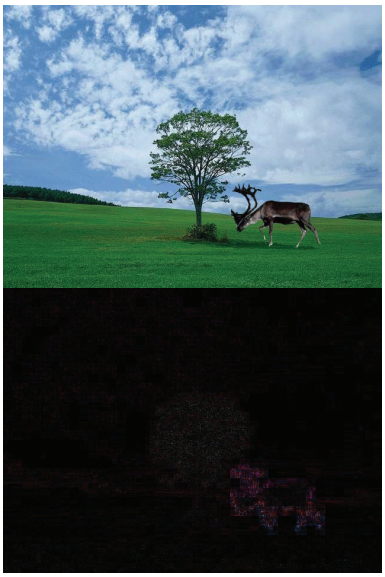

Fig. 1: Forged image and it's ELA image

As we can see in **Fig. 1** the regions that appear to be brighter in ELA are considered to be forged.

C. CNN – Convolutional Neural Network

Convolutional neural network (CNN) [9] is a feedforward neural network architecture for deep learning which learns from data itself. The term convolution is used to describe the mathematical phenomenon of feeding the output of one function into another function repeatedly. It was originally developed for handwriting detection but it has also proven to be helpful in the areas of image classification, identification, segmentation, and detection. It tries to replicated the human visual nervous system. CNN is very useful when tackling data

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TABLE I: Datasets

Dataset Name	Configuration	Size of Images (in pixels)	Characteristics
CASIA v2.0	7,200 pristine images and 5,123 tampered images	Between 320 x 240 to 800 x 600	It contains blurring of the tampered image set besides the splicing
MICC-F220	110 authentic and 110 tampered images	Between 722 x 240 and 800 x 600	The tampered region represents 1.2% of the entire image
MICC-F2000	1300 authentic and 700 tampered images	2048 x 1536	The tampered region comprises of 1.12% of the entire image

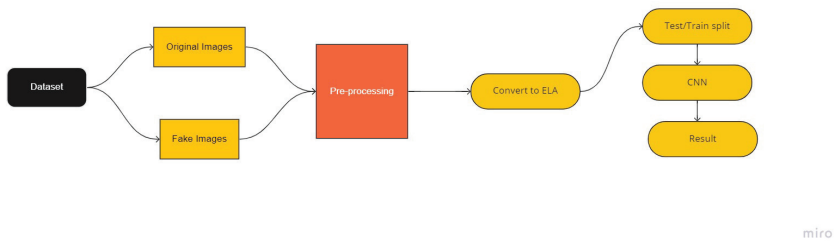

Fig. 2: Proposed Model

with large number of dimensions. CNN effectively reduces the dimensionality while maintaining the relevant and important features of the image. The CNN architecture comprises of three parts: a convolution layer, a fully connected layer and a pooling layer. Features are extracted from the convolution; the fully connected layer is used to get the desired output which is similar to its use in traditional neural networks; and the pooling layer is responsible for the dimensionality reduction and also prevents overfitting. We can calibrate multiple filters, with values that are both variable and trainable, in the convolution layer of CNN to acquire various feature maps so that it becomes easier to classify the filtered output.

The initial CNN architecture employed intricate models with numerous layers, but the objective was to devise a simplified CNN model that could efficiently detect image manipulation in comparison to existing models.

Fig. 2 Shows the overall structure of the proposed model which is elaborated in detail in this section.

D. Feature Extraction

The proposed model is made up of a total of five convolutional layers with a max pool layer of size 2 x 2 being added after every convolutional layer. Dimensionality reduction of feature map and the selection of the maximum element from a specific region is the enforced by the max pool layer. The initial image input size is set to (224×224×3), and the CNNs utilize filters to recognize features, including edges, across the whole input image.

In all the five layers of the proposed model, the convolution layer has a filter size of (3, 3) and is supported by a RELU activation function (equation 1) [10], followed by a max pooling layer of size (2, 2). The first layer receives an input of size (224, 224, 3). The distinguishing factor between all the layers is the number of feature maps present in each layer, gradually increasing, with the first one having 16 features maps followed by the second one having 32, the third one contains 64 feature maps and the fourth and fifth layer contain 128 and 256 feature maps respectively.

$$Relu(x) \quad = max(0, x) \tag{1}$$

Fig. 3 shows the detailed architecture of CNN used in the proposed method.

E. Classification

Moving on to the final stage of the proposed model, it involves the classification of the input image. The classification part consists of a global average pooling layer and the output that was obtained from the final layer of the convolutional process is fed into it. This is followed by a fully connected layer that receives as input the vectors acquired from final feature maps produced after the pooling process using SoftMax [11] Activation function (equation 2). Binary cross entropy function [12] was used as the loss function (equation 3) and Adam [13] as the optimizer for quicker computation. Eventually, the authenticity of the input image can be determined by detecting whether it is a forgery or an original image.

$$\sigma(y_i) = \left(\frac{e^{y_i}}{\sum_j e^{y_j}} \right) j = 1, ..., n \tag{2}$$

here, σ represents the SoftMax value and yi represents the ith value.

$$BCE = (-(y \log(p) + (1 - y) \log(1 - p))) \, j = 1, ..., n \tag{3}$$

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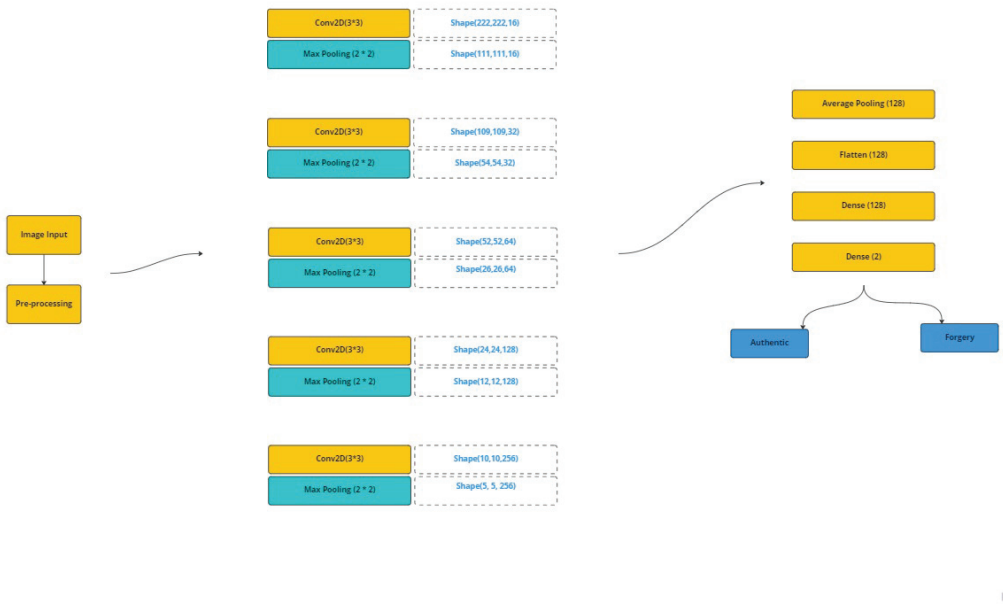

Fig. 3: CNN Architecture

where, y is the binary indicator (0 or 1) if class label c is the correct classification for observation o and p is the predicted probability observation o is of class c

V. Experimentation and Results

For our study, we used a device with 8GB RAM, Intel CPU, and no GPU. For our proposed model, we have evaluated the performance using the current training setup of 100 epochs with 32 batch size and a learning rate of 0.0001. We tested ResNet50, VGG16, and VGG19 architectures using a training setup of 10 epochs, 32 batch size, and a learning rate of 0.0001, We also varied the learning rate, batch size, and number of epochs to determine their effect on model accuracy.

A. Comparative Analysis

1) Based on statistical features of block DCT coefficients: The test image provided for this study is first transformed into the YCbCr colour space. Human eyes are better able to distinguish between the luminance and chroma components and most of the indications of tampering are concealed in the chroma components. The chroma components which are Y, Cr and Cb produce features which when combined together produce a terminal feature vector. 2D-DCT operation was carried out on each of the 8x8 size pixel blocks which are not overlapping with each other. This in used to create a sub-image. In order to generate the concluding feature vector we use the three feature vectors generated from these 3 sub-images(Y, Cr and Ch). The distinction between the authentic and spliced image feature vectors, distinction between the authentic and copy-moved image feature vectors derived from the test image is thought to be a distinguishing factor and can be used to classify images. Classification of images is done using SVM [14].

2) Based on LBP-DCT and SVM Classifier: The input image dataset is converted into Gray, Y, Cb, Cr colour space. The output of the first phase is then divided into blocks and is fed into the block DCT for further processing. The magnitude component of the DCT coefficient is extracted and fed into LBP operator. After this the block division of LBP image takes place and the mean featured are extracted from it which are later fed into the SVM classification model. LBP suppresses other pixels by considering only nearby pixels with values strictly more than the candidate pixel value. LBP is used after and not vice-versa due to reduced accuracy because LBP removes some of the forging artefacts that DCT may record. DCT can detect small changes, some of which LBP can later improve. As a result, this enhancement has a significant impact on mean-based features, making them more visible [15].

3) : The proposed method was also compared against three trained CNN architectures – VGG16, VGG19 [16] and RESNET50 [17] and the results of the same are elaborated further.

B. Evaluation Metrics

The metrics used for evaluation of the results are – accuracy, precision, and recall.

Accuracy: It is the measure of number of times the model correctly identifies the classes in comparison to total number of classifications made.

$$Accuracy = \frac{(TP + TN)}{(TP + FP + FN + TN} \tag{4}$$

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TABLE II: Results

Algorithm/Dataset	CASIA V2.0			MICC-F220			MICC-F2000		
	Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall
DCT+SVM	0.97	0.95	0.92	0.86	0.83	0.78	0.93	0.88	0.90
DCT+LBP+SVM	0.96	0.94	0.92	0.92	0.86	0.91	0.94	0.92	0.89
ELA+VGG16	0.72	0.66	0.67	0.87	0.82	0.88	0.89	0.88	0.90
ELA+VGG19	0.74	0.69	0.72	0.88	0.89	0.85	0.90	0.89	0.88
ELA+RESNET50	0.60	0.59	0.55	0.70	0.60	0.85	0.76	0.65	0.80
ELA+CNN (Proposed)	0.98	0.95	0.92	0.97	0.93	0.91	0.98	0.96	0.94

Precision: It is the measure of how many times the model correctly identifies a class A (positive) compared to total number of items which are class A.

$$Precision = \frac{TP}{TP + FP} \tag{5}$$

Recall: it is a measure of how many times a class A(positive) is correctly identified when compared to all the correct classification made by the model

$$Recall = \frac{TP}{TP + FN} \tag{6}$$

C. Results

Different types of statistical and deep learning methods were tested along with our proposed method on three different datasets – CASIA V2.0, MICC-F220, MICC-F2000. The results can be used to deduce the most efficient method which works well across all the image datasets. **Table 2** shows the metrics obtained upon testing. According to the results obtained in this research paper, the CNN model used with ELA achieved high validation accuracy for forgery detection on different datasets, with 98.25% for CasiaV2, 97.32% for MICC F-220, and 98.67% for MICC F-2000. Moreover, the proposed method outperformed commonly used statistical methods such as DCT and LBP+DCT, as well as popular pretrained models including VGG16, VGG19, and ResNet50. Its significant to point out that the statistical methods were able to outperform the other deep learning methods on the CASIA V2.0 dataset. The proposed method obtained lowest accuracy on the MICC-F220 dataset even when this dataset is the smallest of the three. This can be attributed to the fact that this dataset contains images of variable sizes. Whereas, the lowest average accuracy was obtained on the CASIA V2.0 dataset which can be due to its larger size and varying image resolutions. This points out the need for having better and updated forged image datasets so that researches can thoroughly test their methods. The CNN architecture introduced in this paper was able to work better with ELA than other well-established architectures. Notably, the testing accuracy for most models was consistent with the validation accuracy. These results suggest that proposed CNN model with ELA pre-processing could be an effective approach for forgery detection in various scenarios.

VI. Conclusion and Future Work

Image and video tampering has become a nuisance in society today. With increasing amount of data available online and the technological advancements taking place, this problem is only going to increase. The topic of forgery detection requires the amount of attention it is receiving and it needs to be updates with the ever-evolving tools and software. In this paper we proposed a methodology to solve this problem by employing ELA and CNN together. The results produced by the approach were able to achieve high accuracy across multiple databases available with low computational power required. Our approach when evaluated besides some state-of-the-art-methods perform better on selected datasets. The paper elaborately outlines the use of ELA for feature extraction and how it makes the process of forgery detection easy by providing a ball-park area where the image has the highest possibility of being manipulated. The CNN algorithm has proven it's mettle in the field of processing pixel data and the paper also affirms the power of CNN algorithm to process and classify images. In future, this methodology can be extended to being used in the field of video tampering detection, image/video segmentation or detection of multiple types of tampering methods. In a world where the amount data being exchanged is increasing at an exponential rate, the need to make methods more robust and time efficient always exists. More effort can be invested in building better image datasets which include varied samples to thoroughly test detection methodologies. As the precision of tools and software is increasing in making the forged images, it becomes important for people to research and tackle the different use cases being generated every day.

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